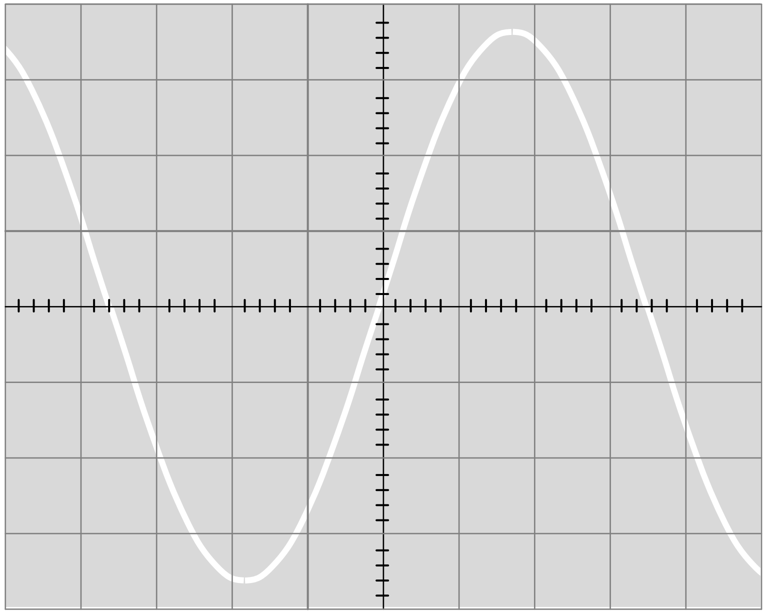
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*	(i)	Rms pd is required or $V_{\text{rms}} = \frac{V}{\sqrt{2}}$ (1) Substitution into $P = \frac{V^2}{R}$ or equivalent e.g. calculating I, $P = IV$ (1)	2			2		
		(ii)	$\frac{18}{5}$ =3.6 squares seen or implied (1) $T = \frac{1}{f}$ used (0.0143 or 14.3 ms) (1) Dividing period by 2 ms (7.1 squares) (1) Period correct in diagram **(ecf)** (1) **accept any phase,** Amplitude correct in diagram **(ecf)** (1) **at least one cycle** 	1	1 1 1 1		5	2	

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(b)	(i)	Reactance of inductor equal and opposite to capacitance (or pds equal and opposite) (1) Minimum impedance (or all pd across resistor) (1) Maximum current (1)	3			3		
		(ii)	Algebra leading to $f = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}$ (1) Answer = 130 kHz (1)		2		2	2	
		(iii)	X_L = 65 000 [Ω] or X_C = 10 500 [Ω] (1) Substitution into $Z = \sqrt{(X_L - X_C)^2 + R^2}$ = 55 000 Ω (1) Use of $I = \frac{V}{Z}$ leading to Answer = 91 μA (**ecf**) (1)		3		3	3	
		(iv)	Smallest capacitor used (or trial and error) (1) Q factor calculated (533) (or other valid method of getting V) (1) Pd across C or L = $Q \times 5$ (2 700 V) (1) Peak pd this is $\times\sqrt{2}$ (3 800 V) (1) Student is wrong, it's considerably larger (valid conclusion) (1)			5	5	3	
			Question 6 total	**6**	**9**	**5**	**20**	**10**	**0**